Econ 440: Experimental Design

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1 Introduction

In these notes, we begin our study of different approaches or Research Designs to estimate causal effects, in particular, the population Average Treatment Effects (ATE) and the Average Treatment Effect on the Treated (ATT). We start with the Gold Standard for estimation of causal effects: Experimental Designs.

Assumption The Key Assumption in an Experimental Design is that T is *independent* of $Y(1)$ and $Y(0)$. We write this as:

$$T \perp\!\!\!\perp \{Y(1), Y(0)\}$$

Notes:

- This assumption holds if T is randomly assigned (RA).
  - For example, assign T by flipping a coin: $T_i = \begin{cases} 1 \text{ if heads} \\ 0 \text{ if tails} \end{cases}$
  - In practice, T is randomly assigned by some other means (e.g., by using a random generating number).
- If T is RA, by construction, it is *uncorrelated* to any confounder.

Question: How do we use this assumption to identify and estimate the ATE?

2 Aside: Review of Statistical Independence

Let X and Y be two random variables.

Definition: X and Y are *independent* if:

$$\Pr(X = x \text{ and } Y = y) = \Pr(X = x) \cdot \Pr(Y = y),$$

if the variables X and Y are discrete; or:

$$f(x, y) = f_X(x) \cdot f_Y(y),$$

if the variables X and Y are continuous, where $f(x, y)$ is the joint probability density function (pdf) of X and Y, $f_X(x)$ is the pdf of X, and $f_Y(y)$ is the pdf of Y.

Two Key Properties of Independence:

1. If X and Y are independent $\Longrightarrow Cov(X, Y) = 0$ and $Corr(X, Y) = 0$.

2. If X and Y are independent $\Longrightarrow E(Y|X) = E(Y)$

Example: A coin is flipped.

Let $Y = \begin{cases} 1 \text{ if heads} \\ 0 \text{ if tails} \end{cases}$

Let $f(y) = Pr(Y = y)$ be the probability (mass) function of Y, such that:

$$f(y) = Pr(Y = y) = \begin{cases} \frac{1}{2} \text{ if } y = 1 \\ \frac{1}{2} \text{ if } y = 0 \end{cases}$$

First, we find $E(Y)$, that is, the expectation (or true, mathematical, or population average) of Y. From our Statistics Review Notes, we have:

$$E(Y) = \sum_{y=0}^{1} y \cdot f(y) = \sum_{y=0}^{1} y \cdot Pr(Y = y) = 0 \cdot Pr(Y = 0) + 1 \cdot Pr(Y = 1) = Pr(Y = 1) = \frac{1}{2}$$

$\Longrightarrow \boxed{E(Y) = \frac{1}{2}}$

(Aside: Note that since $Y = \{0, 1\} \Longrightarrow \boxed{E(Y) = \Pr(Y = 1)}$. This is true for *any* binary variable.)

Now, define the random variable $X = \begin{cases} 1 \text{ if a man flips the coin} \\ 0 \text{ if a woman flips the coin} \end{cases}$

Clearly, X and Y are independent!

$Pr(Y = 1) = Pr(heads) = \frac{1}{2}$ regardless of X, i.e., regardless of who flips the coin (a woman or a man). Therefore:

$E(Y) = E[Y|X = 1] = E[Y|X = 0] = \frac{1}{2}$

In words, the expected value of Y is the same in the population, or if a man flips the coin, or if a woman flips the coin.

3 Identification of the ATE under RA

3.1 How does RA solve the Fundamental Problem of Causal Inference (FPCI)?

Remember from our Potential Outcome Framework Notes equation (3), that we can write the population Average Treatment effects, $ATE = E[Y_i(1) - Y_i(0)]$, as:

$$\begin{aligned} ATE = Pr(T_i = 0)E[Y_i(1)|T_i = 0] + Pr(T_i = 1)E[Y_i|T_i = 1] \\ -Pr(T_i = 0)E[Y_i|T_i = 0] - Pr(T_i = 1)E[Y_i(0)|T_i = 1] \end{aligned} \tag{1}$$

How do we estimate our missing counterfactuals $E[Y_i(1)|T_i = 0]$ and $E[Y_i(0)|T_i = 1]$ needed to estimate the ATE under RA of the treatment? How does RA solve the FPCI?

When T is randomly assigned, T is *independent* of the potential outcomes $Y(0)$ and $Y(1)$, and the following is true (by the second key property of independence in Section 2):

$$E[Y(1)|T = 0] = E[Y(1)|T = 1] = E[Y|T = 1]; \text{ and}$$

$$E[Y(0)|T = 1] = E[Y(0)|T = 0] = E[Y|T = 0]$$

After plugging the two equalities above into equation (1), we obtain:

$$ATE = E[Y|T = 1] - E[Y|T = 0] \tag{2}$$

Therefore, under RA, the ATE is *identified*, i.e., it can be written as a function of the *observed* variables (our data): T_i and $Y_i \implies$ We can estimate it!

In words, if T is RA, equation (2) shows that the ATE equals the expected value of Y for the treated group ($T = 1$) minus the expected value of Y for the control group ($T = 0$).

- Note: T_i is independent of the *potential* outcomes $Y_i(1)$ and $Y_i(0)$, but is not independent of the *observed* outcome $Y_i = Y_i^{obs} = T_i Y_i(1) + (1 - T_i) Y_i(0) = \begin{cases} Y_i(1) \text{ if } T_i = 1 \\ Y_i(0) \text{ if } T_i = 0 \end{cases}$

(Note how Y_i is a function of T_i, so T_i and Y_i are *not* independent.)

Therefore, for example,

$$E[Y(1)|T = 1] = E[Y(1)|T = 0] \qquad \text{(by independence)},$$

but, in general,

$$E[Y|T = 1] \neq E[Y|T = 0] \quad \text{(as } T \text{ and } Y \text{ are not independent)}.$$

Hence, the importance of distinguishing between *potential* and *observed* outcomes!

3.2 Why Does Random Assignment (RA) Work? Intuition

If T is Randomly Assigned (RA), then the treatment ($T = 1$) and control ($T = 0$) groups are *"comparable"*, i.e., they share the *same* characteristics (on average), so it is a *ceteris paribus* comparison.

- Let $X_1, X_2, ..., X_K$ be a set of K *covariates*, that is, observed variables measured *before* treatment is received (also called *"pre-treatment"* variables). Examples:
  - ability (I.Q.), parents' income, age, gender, race, etc. *before* attending college.
  - education, age, experience, employment status, income, etc. *before* enrolling in a training program.

- If T is RA $\Longrightarrow$ T is also independent of **any** *observed* and *unobserved* pre-treatment variables.

  In particular, focusing on the observed covariates, for *any* observed covariate X_k (e.g., gender), we have:

$$T \perp\!\!\!\perp X_k \Longrightarrow E[X_k] = E[X_k|T = 1] = E[X_k|T = 0]$$

Hence, if we could calculate $E[X_k|T = 1]$ and $E[X_k|T = 0]$ for every covariate $k = 1, ..., K$ (remember, these are expectations or population averages, so they are unknown) and do a table of *population* average characteristics of the treatment ($T = 1$) and Control ($T = 0$) groups, they would be equal.

- Example: Let T be "college attendance", and assume it was RA.
  Table of *population* average characteristics of treatment and control groups:

$\Longrightarrow$ Population Average Characteristics are *"balanced"* between treatment ($T = 1$) and Control ($T = 0$) groups

$\Longrightarrow$ It is a *ceteris paribus* comparison: Nothing else changes, only T

$\Longrightarrow$ We get a *causal average effect* of T on Y by comparing the average outcome of the treatment and control groups: $ATE = E[Y|T = 1] - E[Y|T = 0]$!

Importantly, while the discussion above explained the idea in terms of observed covariates, if T is RA, **both** *observed and unobserved pre-treatment variables are balanced between treatment ($T = 1$) and control ($T = 0$) groups!*

For example, for *any* unobserved confounder U (e.g., innate ability), we have:

$$\text{If } T \text{ is RA} \Longrightarrow T \perp\!\!\!\perp U \Longrightarrow E[U] = E[U|T = 1] = E[U|T = 0]$$

3.3 Population Expectations versus Sample Averages

The discussion above focused on expectations (i.e., population, true, "infinite sample", or mathematical averages). As we know, those quantities are (fixed but) unknown to us. Thus, the result above that "for any observed or unobserved pre-treatment variable X_k $E[X_k|T = 1] = E[X_k|T = 0]$ if T is RA," holds for the population average or expectation.

However, in a given finite sample (i.e., in practice), the *sample* averages of observed (and unobserved, if we could see them) variables X_k between treatment and control groups *in our sample* will *not* be exactly equal because of *sample variance.*

In principle, however, if T is indeed RA *and* we have a large enough sample size, we would expect to find that $E[X_k|T=1]$ and $E[X_k|T=0]$ are *statistically equal* for all our covariates X_k.

- That is, we would expect that if we do a test of equality of means for any covariate X_k between the treatment and control groups, we should not reject the null hypothesis that the means $E[X_k|T=1]$ and $E[X_k|T=0]$ are equal.

- Example/Analogy:

  - Suppose we flip a coin. Let $Y = \begin{cases} 1 \text{ if heads} \\ 0 \text{ if tails} \end{cases} \implies E[Y] = \Pr(Y=1) = \frac{1}{2}$

  - If we flip the coin n times, say $n = 10$, then the number of heads in 10 flips may be 3, 6, 7, or any number between 0 and 10. It is not always 5!

    $\implies \overline{Y} = \frac{\sum_{i=1}^{10} Y_i}{10}$ is **not** always $\frac{1}{2}$

  - However, in principle, if we test the null hypothesis that H_0: $E(Y) = \frac{1}{2}$, versus the alternative hypothesis that H_a: $E(Y) \neq \frac{1}{2}$, we would expect not be able to reject the null hypothesis H_0.

- In some cases, though, we may get an "unlucky" draw: e.g., we may flip a coin 10 times and get only 1 head $\Longrightarrow$ we may reject H_0: $E(Y) = \frac{1}{2}$, *even if it is true* (the so-called "Type I Error" in statistics).

- To minimize the chances of getting an "unlucky" (or non-representative) draw, it is important to work with *large samples*:

  * By the *Law of Large Numbers (LLN)*, as $n \to \infty$ (say, as our sample size gets larger), the probability that $\overline{Y}$ is arbitrarily close to $E(Y)$ approaches one.

  * Intuitively, as we flip the coin many more times ($n \to \infty$), $\overline{Y} = \frac{\sum_{i=1}^{n} Y_i}{n} \to \frac{1}{2}$.

  * Hence, the larger our sample, the lower the chances of rejecting the null hypothesis that H_0: $E(Y) = \frac{1}{2}$ when it is indeed true because we would expect the sample average $\overline{Y}$ to be closer to $E(Y) = \frac{1}{2}$.

- **Importantly,** similar ideas apply to our result that, if T is RA, for any observed (or unobserved) covariate X_k: $E[X_k|T = 1] = E[X_k|T = 0]$.

• As in the "flip-a-coin" example above:

- In some cases, *even if T is randomly assigned*, we may get an "unlucky" draw and have that, for some covariates X_k, we *reject* the null hypothesis $H_0 : E[X_k|T = 1] = E[X_k|T = 0]$. As in the example above, this is more likely the smaller our sample size.

- Therefore, it is always better to work with large samples, since by the *LNN*, for **all** covariates X we will have that:

  * The *sample* average of X for the *treated* individuals *in our sample* will be closer to its population average $E[X|T = 1]$; *and*

  * The *sample* average of X for the *control* individuals *in our sample* will be closer to its population average $E[X|T = 0]$.

  * Therefore, if T is RA, we will be less likely to reject the null hypothesis H_0: $E[X_k|T = 1] = E[X_k|T = 0]$ and, <u>*more importantly*</u>, it will be more likely that the average characteristics of the treatment and control individuals **in our sample** are balanced (rather than "only in the population"), thus making the argument for estimating a causal effect stronger.

- Note: Since we always work with finite samples (rather than "infinite samples"), even if T is RA, the average characteristics of the treated ($T = 1$) and control ($T = 0$) groups *in our sample* (i.e., their *sample averages*) will not be exactly equal to each other (although, as discussed above, they will be close). Because of this, it is common to adjust for these small imbalances in the covariates when estimating average causal effects (e.g., by using OLS), as we discuss below.

- Note: The above discussion was done in the context of observed variables. However, the same holds for *unobserved* variables.

  - For example, even if T is RA, the average characteristics of unobserved variables between treated ($T = 1$) and control ($T = 0$) groups in a particular *sample* may differ (if we could see them), but we would expect them to be closer to each other the larger our sample (by the LLN).

<u>In sum:</u>

- It is important to work with large samples (when possible) to let the randomization of T *and* the LLN ensure that the observed and unobserved *pre-treatment* characteristics are balanced between the treatment ($T = 1$) and control ($T = 0$) groups *in our sample.*

- Even in an experiment, it is always important to analyze the balance of the covariates between our treatment and control groups (i.e., analyze their average characteristics) to check if the Random Assignment of T was successful in balancing the covariates.

  - If the *observed* covariates are balanced $\Longrightarrow$ That gives us confidence that the RA of T (and the LLN) is also balancing the *unobserved* pre-treatment variables.

    $\Longrightarrow$ This would give us confidence that we are making a *ceteris paribus* comparison, and therefore, that we are indeed estimating a *causal average effect.*

Final Note: Remember that by the way Hypothesis Testing is done in statistics (as you have seen in your statistics/econometrics classes), in a finite sample, when we perform a test at the 5 percent significance level test ($\alpha = 0.05$), we would expect to reject the null hypothesis *even if it is true* with a 5% probability. In other words, there is a 5% probability of a Type I error (rejecting the null hypothesis when it is true).

- For our purposes, this would imply that if we have 100 covariates and T was RA, we may expect to (incorrectly) reject the null hypothesis that $E[X|T=1] = E[X|T=0]$ for 5 of those covariates.

4 Estimation of the ATE and ATT under RA

4.1 How do we estimate the ATE?

- Assume we have a random sample $i = 1, ..., n$ from the population: $\{T_i, Y_i, X_i\}_{i=1}^n$, where X_i is a $K \times 1$ vector of characteristics of individual i.
- From equation (2), we have that: $ATE = E[Y|T=1] - E[Y|T=0]$.
- How do we estimate the ATE? As commonly done in econometrics, we use *"sample analogue estimators"*: use *sample* means instead of *population* means.
  - By the LLN, *sample* averages are good estimators of *population* averages.
  - Therefore, we estimate the ATE as: $\widehat{ATE}$ = sample mean in treated group ($T = 1$) - sample mean in control group ($T = 0$)

    or, formally,

$$\widehat{ATE} = \frac{\sum_{i=1}^n Y_i T_i}{\sum_{i=1}^n T_i} - \frac{\sum_{i=1}^n Y_i (1 - T_i)}{\sum_{i=1}^n (1 - T_i)} \tag{3}$$

  - *Notation:* We use a hat in a parameter to denote its estimator. For example, given a parameter α, we denote its estimator as $\widehat{\alpha}$.
- An easy way to calculate the estimator $\widehat{ATE}$ in (3) is by running the regression:

$$Y_i = a + bT_i + \epsilon_i, \tag{4}$$

  with $E[\epsilon_i] = 0$. Under RA of T, in this regression: $b = ATE$ and $a = E[Y|T=0] =$ = expected outcome for control group.

Why? Given $Y_i = a + bT_i + \epsilon_i$:

$$E[Y_i|T_i = 1] = a + b + E[\epsilon_i|T_i = 1]$$

$$E[Y_i|T_i = 0] = a + 0 = E[\epsilon_i|T_i = 0]$$

$$\implies ATE = E[Y_i|T_i = 1] - E[Y_i|T_i = 0] = b$$

Therefore, the estimated value of b is our estimate of the ATE. Indeed, the OLS estimator of b, $\widehat{b}$, is mathematically equal to $\widehat{ATE}$ in equation (3): $\boxed{\hat{b} = \widehat{ATE}}$

Similarly, the OLS estimator of the intercept a, $\widehat{a}$, equals the sample average of the outcome (Y) for the control group, or:

$$\widehat{a} = \widehat{E}[Y|T = 0] = \frac{\sum_{i=1}^{n} Y_i(1 - T_i)}{\sum_{i=1}^{n}(1 - T_i)}$$

- From OLS, we could get the standard error of $\widehat{b} = \widehat{ATE}$, which we can use to test hypothesis about ATE (e.g., by getting a t-statistic or a p-value), such as: H_0: $b = 0$ versus H_a: $b \neq 0$.

4.1.1 Identification and Estimation of the ATT

So far, our discussion has centered on the population average treatment effect (ATE), what about the average treatment effect on the treated or $ATT = E[Y_i(1) - Y_i(0)|T = 1]$?

- If T is RA, then T is independent of the potential outcomes, or $T \perp\!\!\!\perp \{Y(1), Y(0)\}$. Hence, we have:

  $ATT = E[Y_i(1) - Y_i(0)|T_i = 1] = E[Y_i(1) - Y_i(0)] = \text{ATE}$

  Therefore, if T is RA: $\implies \boxed{ATE = ATT.}$

  As a result, if T is RA: $\implies \boxed{\widehat{ATT} = \widehat{ATE},}$ i.e., our estimator of the ATE is also our estimator of the ATT.

- <u>Intuition:</u> Since T is RA, the treatment ($T_i = 1$) group is "*comparable*" to (i.e., a "statistical clone" of) the *overall* population (say, they have the *same population* average characteristics). Therefore, for any *observed or unobserved* pre-treatment variable X: If T is RA so that $T \perp\!\!\!\perp X \Longrightarrow E[X] = E[X|T = 1] = E[X|T = 0]$.

$\Longrightarrow$ If T is RA: $ATT = ATE$.

4.2 Some Final Notes on Estimation

- Even if T is RA, we can add the covariates to the OLS of the outcome Y on T in equation (4) when estimating the ATE in order to: **(1)** increase precision (i.e., reduce standard errors), and **(2)** adjust for *minor* remaining imbalances in the covariates X that may have occurred by chance (remember: the sample averages of the covariates for the treatment and control groups will not be exactly equal because of sample variance, but they will be relatively close). This is typically done in practice.
  - In particular, given observed covariates $X_1, X_2, ..., X_K$, we can run the regression:
  $$Y_i = a + bT_i + c_1X_{1i} + c_2X_{2i} + ...c_KX_{Ki} + \epsilon_i, \tag{5}$$
  where, as before, the estimate of b will be our estimate of the ATE.
  - <u>Key:</u> The Estimate of b=ATE in equation (5) should <u>not</u> change much with respect to the one from the OLS of Y on only T in equation (4) in this case, since T is RA and is thus independent of X.
    * Otherwise, this may point to RA of T not being successful in balancing the observed covariates (and most likely also not balancing the unobserved pre-treatment variables) $\Longrightarrow$ Our argument for estimating a causal effect is no longer strong (i.e., confounding may *still* be an issue).
- As discussed above, before estimating the ATE, it is always good to check the *covariate balance* between the treatment and control groups:

- Did RA do a good job at balancing the average characteristics of our treatment and control groups? Does $E[X|T=1] = E[X|T=0]$ for all covariates?

- Remember: We need this in order to argue that we have estimated a causal effect (i.e., that "only T changes").

- We could use OLS to check covariate balance between treatment and control groups.

- Finally, remember that RA balances *all pre-treatment* observed and unobserved variables, but **RA does not balance *post-treatment* variables**.

  - Post-treatment variables may be affected by T, and thus, may be intermediate outcomes or mechanisms/channels.

    Example: If who attended college (T) was RA:

    $\Longrightarrow$ RA does <u>not</u> balance employment or experience *after* attending college.

  - As we have discussed, the covariates X in our context are *pre-treatment* variables, i.e., variables measured before treatment (or at baseline).

5 Drawbacks/Limitations of Experiments

As we have discussed, every approach in econometrics has pros and cons, even randomized experiments. Some drawbacks/limitations of Experimental Designs are:

- Some treatments are difficult to randomize (e.g., "attending college").
- Expensive and can take a lot of time.
- There may be ethical issues.
- Results apply only to a particular population and time.
  - For example, results from an experiment to assess the effects of a training program in California in the 1980's may not be informative for training programs in other states (e.g., Florida) today.
- Other usual problems in practice:
  - *Attrition:* some people are "missing" in follow-up surveys or interviews, leading to missing outcomes.
    - It can cause issues if, as is usually the case, the outcomes are not "missing at random". E.g., suppose we only have outcomes for "rich" people, but not for "poor" people. That may bias our results.
  - *Non-compliance:* people may not comply with their assignment (also called "cross-over").
    - For example, some units assigned to enroll in a program do *not* enroll, and some units assigned not to enroll manage to enroll.

6 Aside: Relation to Omitted Variable Bias (OVB)

We relate our discussion in these notes to the issue of OVB, which you saw in Econ 339.

- Suppose T is <u>not</u> RA and the *true* OLS model is:

$$\underbrace{Y_i}_{\text{Earnings}} = a + b \underbrace{T_i}_{\text{College}} + c \underbrace{A_i}_{\text{Ability}} + \epsilon_i, \tag{6}$$

  with $E(\epsilon_i) = 0$ and ϵ_i independent of T_i and A, so $Cov(T_i, \epsilon_i) = Cov(A_i, \epsilon_i) = 0$.

- Also, suppose that A_i is unobserved and correlated to both T_i and Y_i.

- Since A_i is unobserved, we instead *estimate* the model:

$$Y_i = a + bT_i + U_i \tag{7}$$

Now, $Cov(T_i, U_i) \neq 0$ because A_i is in U_i (now, $U_i = cA_i + \epsilon_i$) and $Cov(T_i, A_i) \neq 0$ $\implies Cov(T_i, U_i) \neq 0$.

- Is $\widehat{b}$ in the regression we estimate in (7) an unbiased estimate of b? (Remember: $\widehat{b}$ is an "unbiased" estimator of b if $E(\widehat{b}) = b$.)

- In this case, it is possible to show (using Econ 339 skills) that:

$$E[\widehat{b}] = b + c \cdot \frac{Cov(T, A)}{Var(T)} \neq b \tag{8}$$

$\implies \widehat{b}$ is <u>biased</u> if:

1. $Cov(T, A) \neq 0$, **and**
2. $c \neq 0$

$\implies$ If $Cov(T, A) = 0$ **or** $c = 0 \implies$ <u>No</u> O.V.B.

In other words, not adjusting for A_i is an issue **only** if it affects **both** T_i and Y_i.

- If T is RA $\implies Cov(T, A) = 0 \implies E(\widehat{b}) = b$, so there is no OVB!

- Also, if T is RA, our estimate $\widehat{b}$ should not change much if we add other covariates to the regression $Y_i = a + bT_i + U_i$. For example, if T is RA, our estimate of b will be very similar if we included or not A_i in our regression (if we could observe it), since $Cov(T, A) = 0$ (see equation (8)).

- Note: It is okay if ability (A_i) still affects Y_i, since $Cov(T, A) = 0$ implies $\widehat{b} = b$.
  - Omitting A_i is only an issue if it affects both T and Y.
  - Remember: For a variable to be a confounder (or to lead to OVB), it must affect **both** T and Y.

7 Appendix

This appendix is for general information only (i.e., its material will not appear in problem sets or exams). It explains a way to arrive to the OVB formula in equation (8).

7.1 Properties of the Covariance

- Let X, Y, Z be three random variables. Then:
  1. **Definition:** $Cov(X, Y) = E[(X - \mu_i)(Y - \mu_y)]$, where $\mu_X = E(X)$ and $\mu_Y = E(Y)$
  2. For constants a and b:
  $$Cov(X, aY + bZ) = aCov(X, Y) + bCov(X, Z)$$
  3. $Cov(X, X) = E[(x - \mu_x)^2] = Var(X)$
  4. $Cov(X, a) = 0$

7.2 OVB Formula

To gain some intuition on how to get to the OVB formula in equation (8), we apply the properties of the covariance in section 7.1 to write:

$$E[\hat{b}] = \frac{Cov(T,Y)}{Var(T)} = \frac{Cov(T, \overbrace{a + bT + cA + \epsilon}^{=Y,\text{ i.e. true } Y})}{Var(T)}$$

$$= \frac{\overbrace{Cov(T,a)}^{=0} + b\overbrace{Cov(T,T)}^{=Var(T)} + c \cdot Cov(T,A) + \overbrace{Cov(T,\epsilon)}^{=0}}{Var(T)}$$

$$= b + c \cdot \frac{Cov(T,A)}{Var(T)}$$